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Vectorized Implementation for Linear Regression (Naive and Curious Approach)

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❖ Linear regression

	Feature	Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

House price data

$$\text{Model: } \hat{y} = w_1 x_1 + b$$

$$\text{price} = a * \text{area} + b$$

Features			Label
TV	↕ Radio	↕ Newspaper	↕ Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9

Advertising data

$$\text{Model: } \hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$\text{Sale} = w_1 * TV + w_2 * Radio + w_3 * Newspaper + b$$

Linear Regression

1) Pick a sample (x_1, x_2, x_3, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = w_1 * TV + w_2 * R + w_3 * N + b$$

$$\hat{y} = w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w_1} = 2x_1(\hat{y} - y) \quad \frac{\partial L}{\partial w_3} = 2x_3(\hat{y} - y)$$

$$\frac{\partial L}{\partial w_2} = 2x_2(\hat{y} - y) \quad \frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w_1 = w_1 - \eta \frac{\partial L}{\partial w_1} \quad w_3 = w_3 - \eta \frac{\partial L}{\partial w_3}$$

$$w_2 = w_2 - \eta \frac{\partial L}{\partial w_2} \quad b = b - \eta \frac{\partial L}{\partial b}$$

Features			Label
TV	↕ Radio	↕ Newspaper	↕ Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9

Advertising data

Model

$$\text{Sale} = w_1 * TV + w_2 * Radio + w_3 * Newspaper + b$$

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

1) Pick a sample (x_1, x_2, x_3, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = w_1 * TV + w_2 * R + w_3 * N + b$$

$$\hat{y} = w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w_1} = 2x_1(\hat{y} - y) \quad \frac{\partial L}{\partial w_3} = 2x_3(\hat{y} - y)$$

$$\frac{\partial L}{\partial w_2} = 2x_2(\hat{y} - y) \quad \frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w_1 = w_1 - \eta \frac{\partial L}{\partial w_1} \quad w_3 = w_3 - \eta \frac{\partial L}{\partial w_3}$$

$$w_2 = w_2 - \eta \frac{\partial L}{\partial w_2} \quad b = b - \eta \frac{\partial L}{\partial b}$$

```
1  # compute output and loss
2  def predict(x1, x2, x3, w1, w2, w3, b):
3      return w1*x1 + w2*x2 + w3*x3 + b
4  def compute_loss(y_hat, y):
5      return (y_hat - y)**2
6
7  # compute gradient
8  def compute_gradient_wi(xi, y, y_hat):
9      dl_dwi = 2*xi*(y_hat-y)
10     return dl_dwi
11 def compute_gradient_b(y, y_hat):
12     dl_db = 2*(y_hat-y)
13     return dl_db
14
15 # update weights
16 def update_weight_wi(wi, dl_dwi, lr):
17     wi = wi - lr*dl_dwi
18     return wi
19 def update_weight_b(b, dl_db, lr):
20     b = b - lr*dl_db
21     return b
```

Motivation

Feature		Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

House price data

Features			Label
TV	Radio	Newspaper	Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9

Advertising data

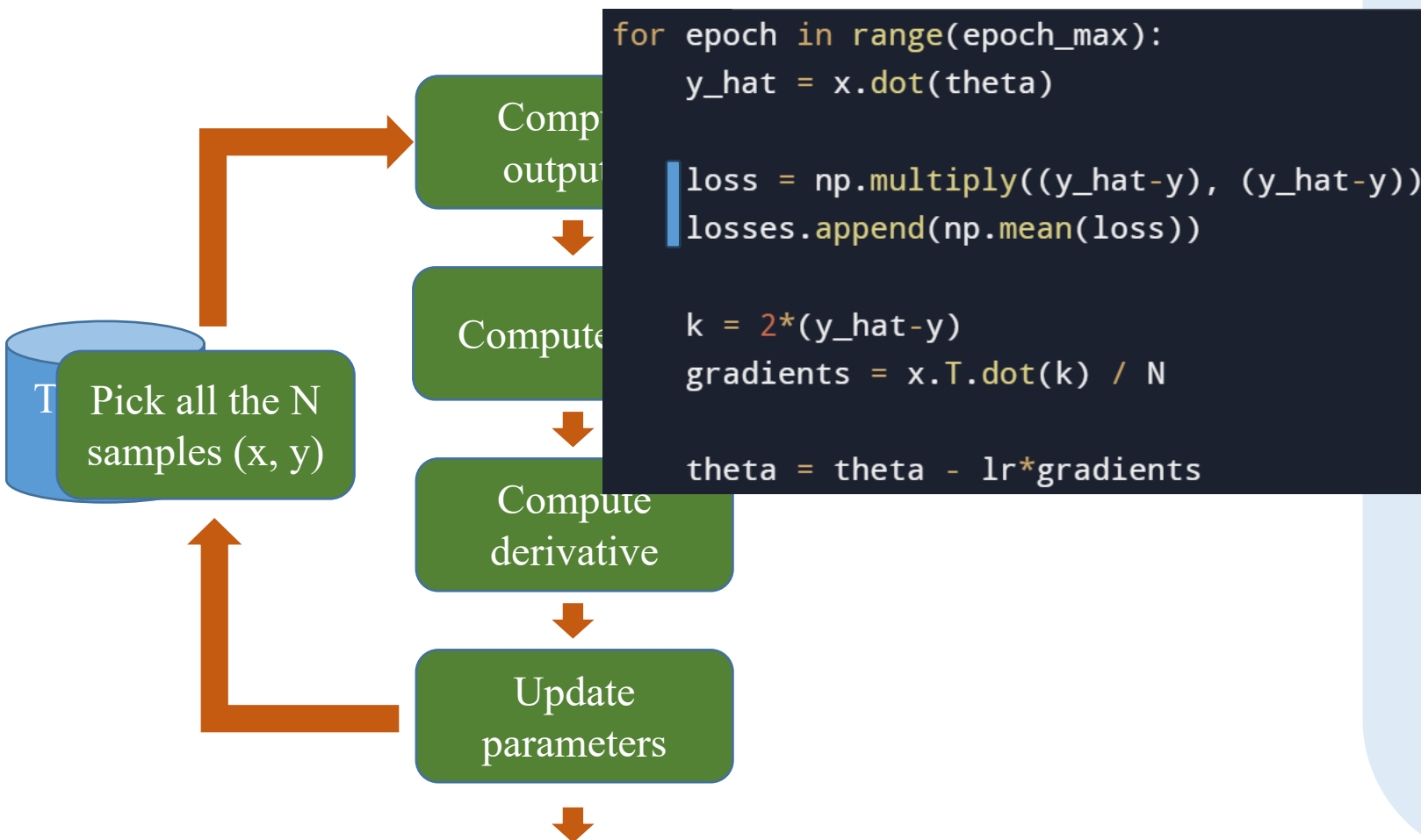
if area=6.0, price=?

if TV=55.0, Radio=34.0,
and Newspaper=62.0,
price=?

Features													Label
crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
0.00632	18	2.31	0	0.538	6.575	65.2	4.09	1	296	15.3	396.9	4.98	24
0.02731	0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.9	9.14	21.6
0.03237	0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
0.06905	0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.9	5.33	36.2
0.08829	12.5	7.87	0	0.524	6.012	66.6	5.5605	5	311	15.2	395.6	12.43	22.9

Boston House Price Data

- ❖ Using vector and matrix for notation
- ❖ Using Numpy for implementation



1) Pick all the N samples from training data

2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

3) Compute loss

$$L = (\hat{y} - y)(\hat{y} - y)^T \frac{1}{N}$$

4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = X^T k$$

5) Update parameters

$$\theta = \theta - \eta \frac{L'_{\theta}}{N}$$

Outline

SECTION 1

1-sample Vectorization

SECTION 2

m-sample Vectorization

SECTION 3

N-sample Vectorization

	Feature	Label
	area	price
	6.7	9.1
	4.6	5.9
	3.5	4.6
	5.5	6.7
	x	y

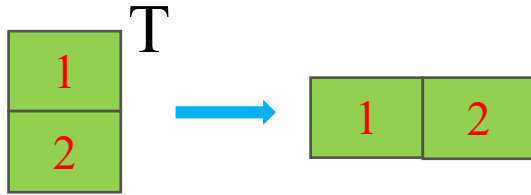
$$\hat{y} = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

Linear Regression (1-samples)

Transpose

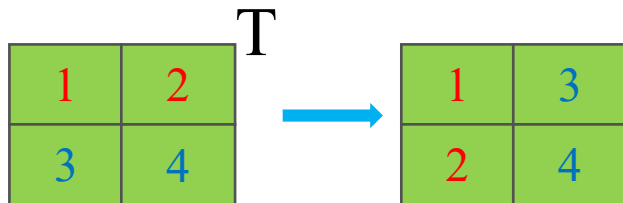
$$\vec{v} = \begin{bmatrix} v_1 \\ \dots \\ v_n \end{bmatrix}$$

$$\vec{v}^T = [v_1 \dots v_n]$$



$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_{11} & \dots & a_{m1} \\ \dots & \dots & \dots \\ a_{1n} & \dots & a_{mn} \end{bmatrix}$$

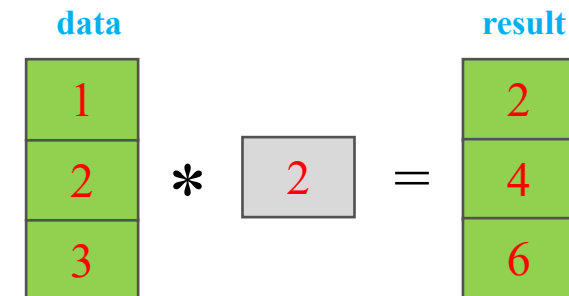


```
2 import numpy as np
3
4 # create data
5 data = np.array([1,2,3])
6 factor = 2
7
8 # broadcasting
9 result_multiplication = data*factor
```

```
[1 2 3]
[2 4 6]
```

Multiply with a number

$$\alpha \vec{u} = \alpha \begin{bmatrix} u_1 \\ \dots \\ u_n \end{bmatrix} = \begin{bmatrix} \alpha u_1 \\ \dots \\ \alpha u_n \end{bmatrix}$$



Linear Regression (1-samples)

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7
x	y

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = wx + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = 2x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

$$\hat{y} = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w \end{bmatrix} \rightarrow \theta^T = [b \quad w]$$

$$\hat{y} = wx + b1 = [b \quad w] \begin{bmatrix} 1 \\ x \end{bmatrix} = \theta^T x$$

dot product

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = wx + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = 2x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

$$\hat{y} = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\hat{y} = \theta^T x$$

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

numbers

What will we do?

Linear Regression (1-samples)

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = wx + b$$

Traditional

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = 2x(\hat{y} - y) \quad \frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \quad b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

$$\hat{y} = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\left\{ \begin{array}{l} \frac{\partial L}{\partial b} = 2(\hat{y} - y) = 2 \times (\hat{y} - y) \times 1 \\ \frac{\partial L}{\partial w} = 2x(\hat{y} - y) = 2 \times (\hat{y} - y) \times x \end{array} \right.$$

$$\begin{bmatrix} 2 \times (\hat{y} - y) \times 1 \\ 2 \times (\hat{y} - y) \times x \end{bmatrix} = \underbrace{2(\hat{y} - y)}_{\text{common factor}} \begin{bmatrix} 1 \\ x \end{bmatrix} = 2(\hat{y} - y)x = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix} = L'_{\theta} \quad \rightarrow \quad L'_{\theta} = 2x(\hat{y} - y)$$

Linear Regression (1-samples)

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = wx + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = 2x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \qquad b = b - \eta \frac{\partial L}{\partial b}$$

Traditional

$$\hat{y} = wx + b$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$L'_{\boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix}$$

$$\left\{ \begin{array}{l} b = b - \eta \frac{\partial L}{\partial b} \\ w = w - \eta \frac{\partial L}{\partial w} \\ \boldsymbol{\theta} \quad \boldsymbol{\theta} \quad L'_{\boldsymbol{\theta}} \end{array} \right.$$

$$\rightarrow \boldsymbol{\theta} = \boldsymbol{\theta} - \eta L'_{\boldsymbol{\theta}}$$

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$\hat{y} = wx + b$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = 2x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta}$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta L'_{\boldsymbol{\theta}}$$

η is learning rate

Vectorized

$$\hat{y} = wx + b$$

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7
x	y

1

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix} = \begin{bmatrix} 1 \\ 6.7 \end{bmatrix}$$

Given $\theta = \begin{bmatrix} b \\ w \end{bmatrix} = \begin{bmatrix} 0.049 \\ -0.34 \end{bmatrix}$
 $\eta = 0.01$

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = \theta^T x = x^T \theta$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

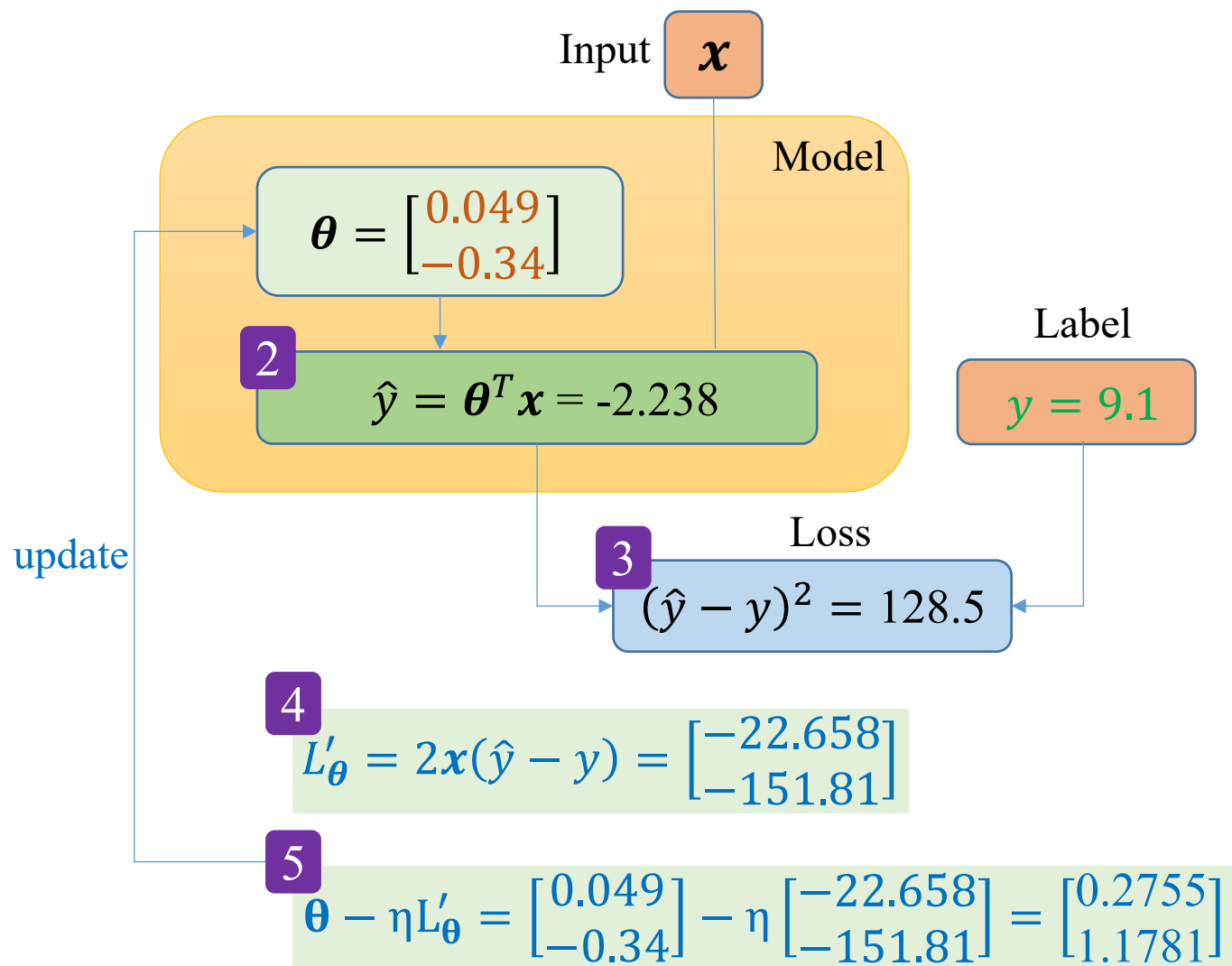
4) Compute derivative

$$L'_\theta = 2x(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta L'_\theta$$

η is learning rate



❖ Implementation (vectorization using numpy)

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = \theta^T x = x^T \theta$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$L'_{\theta} = 2x(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta L'_{\theta}$$

η is learning rate

```
1 import numpy as np
2
3 # forward
4 def predict(x, theta):
5     return x.dot(theta)
6
7 # compute gradient
8 def gradient(y_hat, y, x):
9     dtheta = 2*x*(y_hat-y)
10
11     return dtheta
12
13 # update weights
14 def update_weight(theta, lr, dtheta):
15     dtheta_new = theta - lr*dtheta
16
17     return dtheta_new
```

$$\hat{y} = wx + b$$

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7
x	y

1

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix} = \begin{bmatrix} 1 \\ 3.5 \end{bmatrix}$$

Given $\theta = \begin{bmatrix} b \\ w \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$
 $\eta = 0.01$

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = \theta^T x = x^T \theta$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

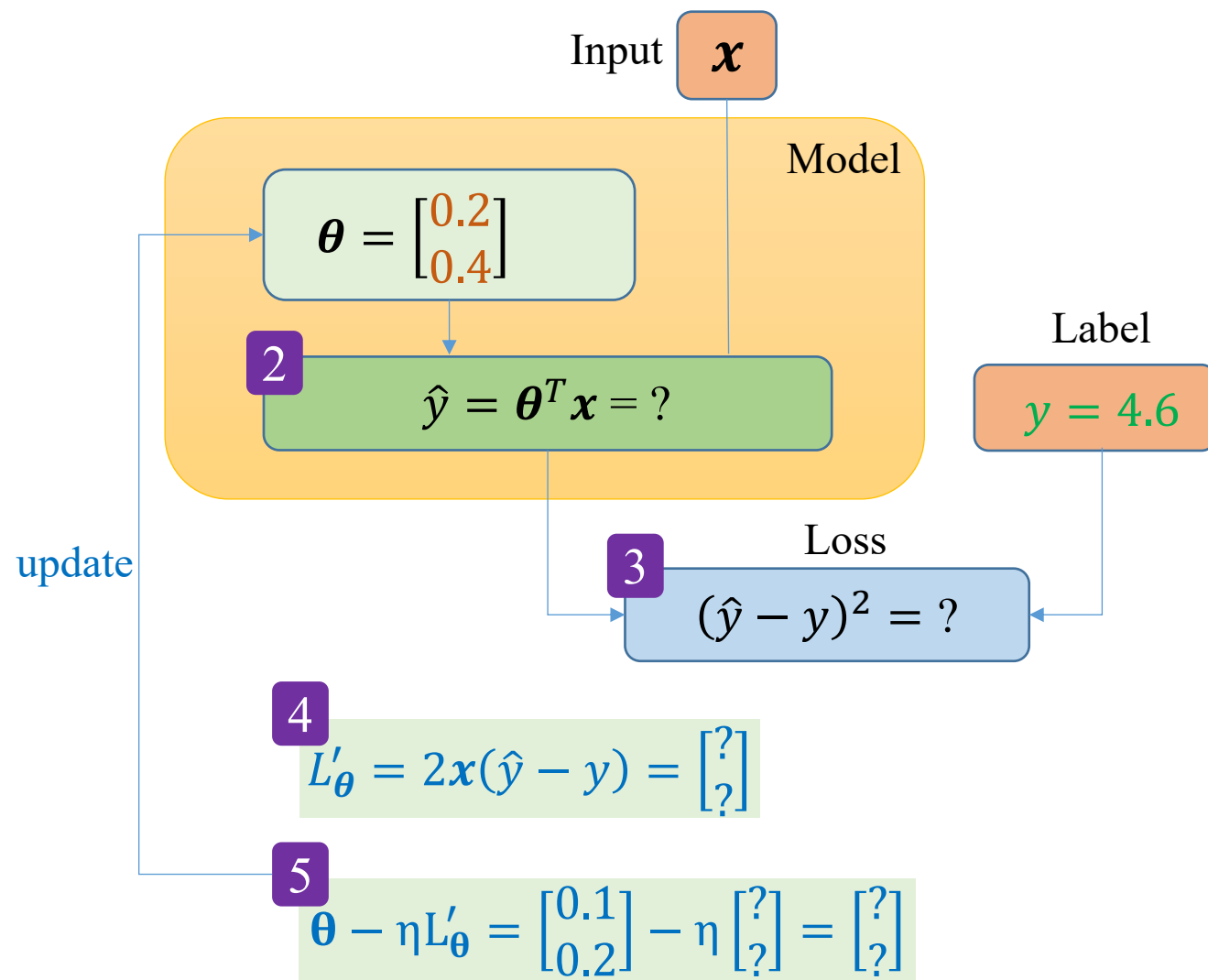
4) Compute derivative

$$L'_{\theta} = 2x(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta L'_{\theta}$$

η is learning rate




```
3 data = np.genfromtxt('data.csv', delimiter=',')
4 N = 4
5
6 areas = data[:, 0].reshape(N, 1)
7 prices = data[:, 1].reshape(N,)
8
9 # vector [1, area]
10 features = np.hstack([np.ones((N,1)), areas])
```

```
1 # forward
2 def predict(x, theta):
3     return x.T.dot(theta)
4
5 # compute gradient
6 def gradient(y_hat, y, x):
7     dtheta = 2*x*(y_hat-y)
8     return dtheta
9
10 # update weights
11 def update_weight(theta, lr, dtheta):
12     dtheta_new = theta - lr*dtheta
13     return dtheta_new
```

```
1 lr = 0.01
2 epoch_max = 10
3
4 # [b, w]
5 theta = np.array([0.049, -0.34])
6
7 for epoch in range(epoch_max):
8     for i in range(N):
9         # get a sample
10         x = features[i,:]
11         y = prices[i]
12
13         # predict y_hat
14         y_hat = predict(x, theta)
15
16         # compute loss
17         loss = (y_hat-y)*(y_hat-y)
18
19         # compute gradient
20         dtheta = gradient(y_hat, y, x)
21
22         # update weights
23         theta = update_weight(theta, lr, dtheta)
```

Advertising Problem

Features			Label
TV	↕ Radio	↕ Newspaper	↕ Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9

Advertising data

if TV=55.0, Radio=34.0,
and Newspaper=62.0,
price=?

$$\hat{y} = w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + b$$

$$x = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

1) Pick a sample (x, y) from training data



2) Compute output $\hat{y}$



$$\hat{y} = \theta^T x = x^T \theta$$

3) Compute loss



$$L = (\hat{y} - y)^2$$

4) Compute derivative



$$L'_\theta = 2x(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta L'_\theta$$

η is learning rate

Outline

SECTION 1

1-sample Vectorization

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

SECTION 2

m-sample Vectorization

Way 1 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

SECTION 3

N-sample Vectorization

Way 2 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

❖ Vectorization

House price data

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Model

$$\text{price} = w * \text{area} + b$$

$$\hat{y} = wx + b$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

Parameter Initialization

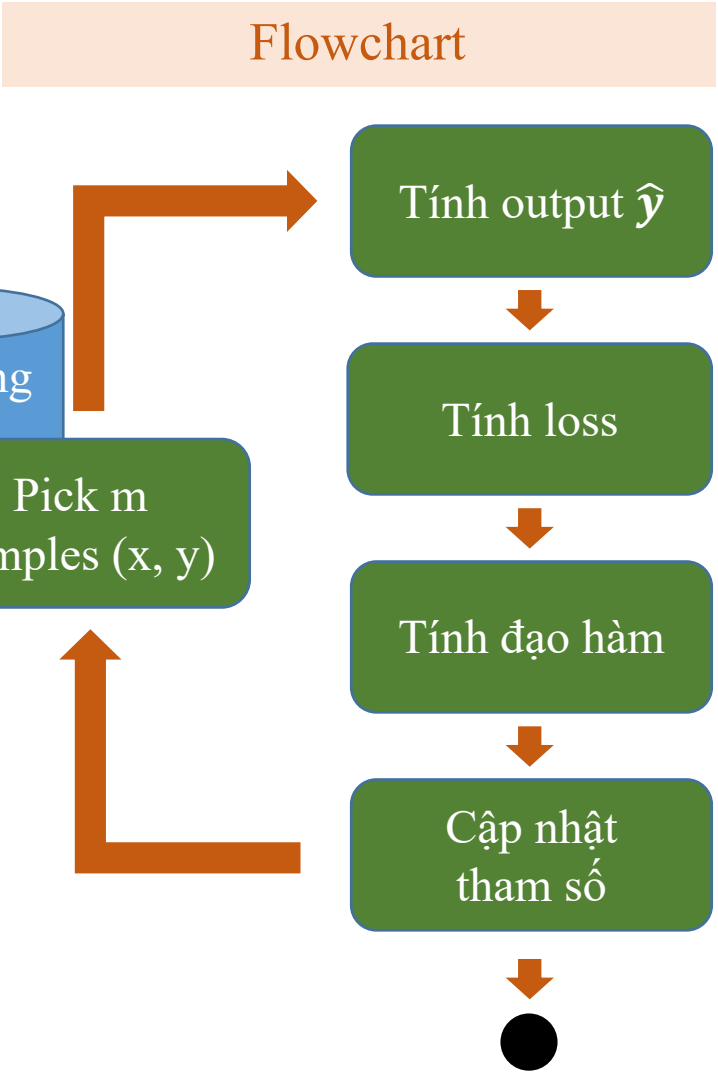
$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

Way 1 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

Way 2 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

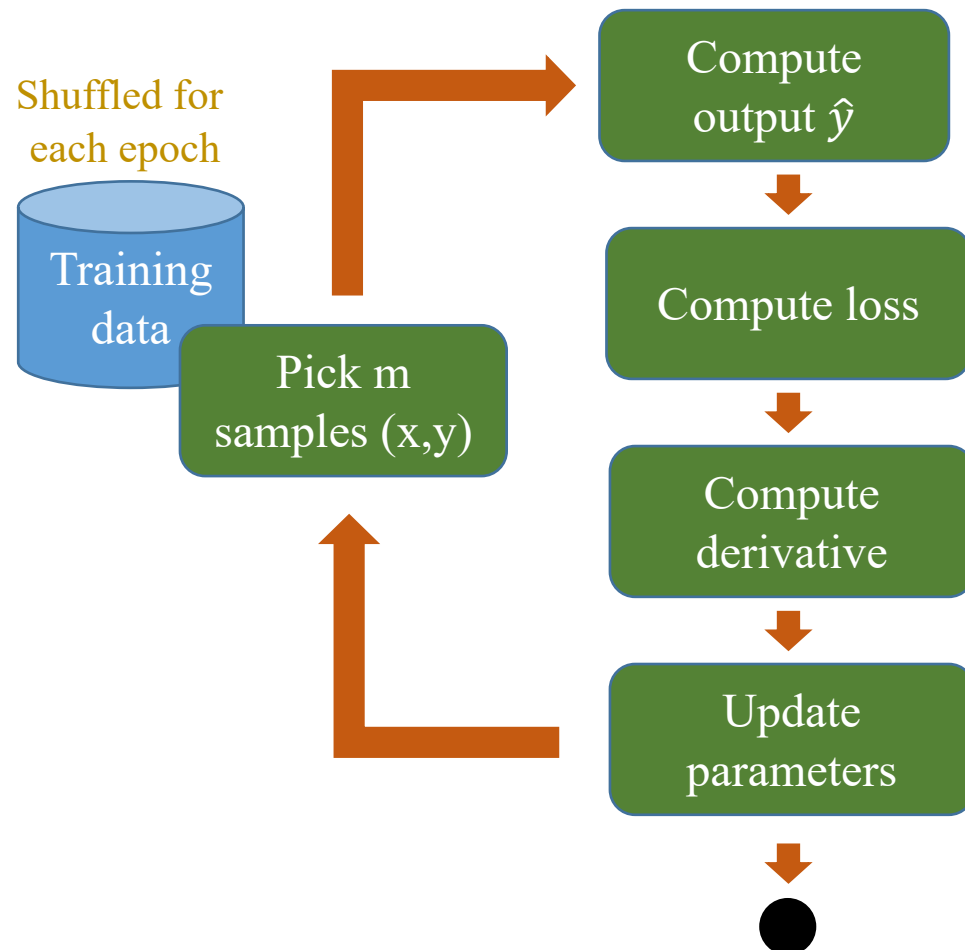


Linear Regression (m-samples)

Feature	Label
area	price
6.7	9.1
4.6	5.9

❖ House price prediction

❖ m-sample training ($1 < m < N$)



1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = wx^{(i)} + b \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_w{}^{(i)} = 2x^{(i)}(\hat{y}^{(i)} - y^{(i)})$$
$$L'_b{}^{(i)} = 2(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$w = w - \eta \frac{\sum_i L'_w{}^{(i)}}{m}$$
$$b = b - \eta \frac{\sum_i L'_b{}^{(i)}}{m}$$

Learning rate η

Linear Regression (m-samples)

Feature		Label
area		price
6.7		9.1
4.6		5.9

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = wx^{(i)} + b \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$\begin{aligned} L'_w{}^{(i)} &= 2x^{(i)}(\hat{y}^{(i)} - y^{(i)}) \\ L'_b{}^{(i)} &= 2(\hat{y}^{(i)} - y^{(i)}) \end{aligned} \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\begin{aligned} w &= w - \eta \frac{\sum_i L'_w{}^{(i)}}{m} \\ b &= b - \eta \frac{\sum_i L'_b{}^{(i)}}{m} \end{aligned} \quad \text{Learning rate } \eta$$

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

Implementation - Differences

Feature		Label	
	area		price
	6.7		9.1
	4.6		5.9

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta}$$

3) Compute loss

$$L = (\hat{y} - y)^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta L'_{\boldsymbol{\theta}}$$

η is learning rate

1) Pick m samples $(\mathbf{x}^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'^{(i)}_{\boldsymbol{\theta}} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'^{(i)}_{\boldsymbol{\theta}}}{m}$$

η is learning rate

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \theta^T x^{(i)} = (x^{(i)})^T \theta \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_\theta = 2x^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\theta = \theta - \eta \frac{\sum_i L'_\theta}{m} \quad \eta \text{ is learning rate}$$

```
1 # vector [x, b]
2 data = np.c_[areas, np.ones((data_size, 1))]
3
4 # init weight
5 lr = 0.01
6 theta = np.array([-0.34, 0.04]) #[w, b]
7
8 # number of epochs
9 epoch_max = 10
10
11 # mini-batch size
12 m = 2
```


Linear Regression

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \theta^T x^{(i)} = (x^{(i)})^T \theta \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\theta}{}^{(i)} = 2x^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\theta = \theta - \eta \frac{\sum_i L'_{\theta}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

```
14 for epoch in range(epoch_max):
15     for j in range(0, data_size, m):
16
17         # some variables
18         sum_of_losses = 0
19         gradients = np.zeros((2,))
20         for index in range(j, j+m):
21             # get mini-batch
22             x_i = data[index]
23             y_i = prices[index]
24
25             # predict y_hat_i
26             y_hat_i = x_i.dot(theta)
27
28             # compute loss
29             l_i = (y_hat_i - y_i)*(y_hat_i - y_i)
30
31             # compute gradient
32             gradient_i = x_i*2*(y_hat_i - y_i)
33
34             # accumulate gradients
35             gradients = gradients + gradient_i
36             sum_of_losses = sum_of_losses + l_i
37
38         # normalize
39         sum_of_losses = sum_of_losses/2
40         gradients = gradients/2
```

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \theta^T x^{(i)} = (x^{(i)})^T \theta \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\theta}{}^{(i)} = 2x^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\theta = \theta - \eta \frac{\sum_i L'_{\theta}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

More vectorization

	Feature	Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

way 1

$$X = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$X = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T X$$

$$= [-0.34 \quad 0.049] \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} = [-2.238 \quad -1.524]$$

Feature		Label
area		price
6.7	4.6	9.1
3.5	5.5	4.6
5.5		6.7

Linear Regression (m-samples)

Way 1.1

Feature		Label
area		price
6.7	4.6	9.1
4.6	3.5	5.9
3.5	5.5	4.6
5.5		6.7

1) Pick m samples $(\mathbf{x}^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$X = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T X = [-2.238 \quad -1.524]$$

$$L = \frac{1}{m} [(\hat{y}^{(0)} - y^{(0)})^2 + (\hat{y}^{(1)} - y^{(1)})^2]$$

$$= \frac{1}{m} [(\hat{y}^{(0)} - y^{(0)}) \quad (\hat{y}^{(1)} - y^{(1)})] \begin{bmatrix} (\hat{y}^{(0)} - y^{(0)}) \\ (\hat{y}^{(1)} - y^{(1)}) \end{bmatrix}$$

$$= \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$= \frac{1}{2} (128.5 + 55.11) = 91.8$$

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$= \begin{bmatrix} -22.676 & -14.848 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

Feature		Label	
	area		price
	6.7		9.1
	4.6		5.9
	3.5		4.6
	5.5		6.7

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$= \begin{bmatrix} -22.676 & -14.848 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} = \begin{bmatrix} -22.676 & -14.848 \\ -22.676 & -14.848 \end{bmatrix}$$

Feature		Label	
	area		price
	6.7		9.1
	4.6		5.9
	3.5		4.6
	5.5		6.7

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

Take a look
and improve

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T) = 2(\hat{\mathbf{y}} - \mathbf{y}^T) \\ = \begin{bmatrix} -22.676 & -14.848 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} = \begin{bmatrix} -22.676 & -14.848 \\ -22.676 & -14.848 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix}$$

Gradient for w from $x^{(i)}$

Gradient for b from $x^{(i)}$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

Take a look
and improve

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T) = 2(\hat{\mathbf{y}} - \mathbf{y}^T) \\ = [-22.676 \quad -14.848]$$

$$\mathbf{x} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} = \begin{bmatrix} -22.676 & -14.848 \\ -22.676 & -14.848 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix}$$

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Take a look
and improve

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

Feature		Label	
	area		price
	6.7		9.1
	4.6		5.9
	3.5		4.6
	5.5		6.7

Linear Regression (m-samples)

Way 1.1

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}{}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

$$\eta = 0.01$$

$$m = 2$$

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m}$$

$$= \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix} - 0.005 \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix} = \begin{bmatrix} 0.761 \\ 0.227 \end{bmatrix}$$

Feature		Label	
	area		price
	6.7		9.1
	4.6		5.9
	3.5		4.6
	5.5		6.7

Linear Regression (m-samples)

2

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < m$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}^{(i)}}{m} \quad \eta \text{ is learning rate}$$

Generalized formula

1) Pick m samples $(\mathbf{x}, \mathbf{y})$ from training data

2) Compute output $\hat{\mathbf{y}}$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X}$$

3) Compute loss

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

4) Tính đạo hàm

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$\odot$ is element-wise
multiplication

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

5) Cập nhật tham số

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m} \quad \eta \text{ is learning rate}$$

More generalized formula

Linear Regression

1) Pick m samples $(\mathbf{x}, \mathbf{y})$ from training data

2) Compute output $\hat{\mathbf{y}}$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X}$$

3) Compute loss

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

4) Compute derivative

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$\odot$ is element-wise
multiplication

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m}$$

η is learning rate

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.genfromtxt('data.csv', delimiter=',')
5 areas = data[:,0]
6 prices = data[:,1:]
7 N = areas.size
8
9 # vector [x, b]^T
10 data = np.vstack([areas, np.ones((N,))])
11
12 # [w, b]
13 theta = np.array([[ -0.34],
14                  [ 0.04]])
15
16 # params
17 lr = 0.01
18 epoch_max = 1
19 m = 2
20
21 # logging
22 losses = []
```

1) Pick m samples $(\mathbf{x}, \mathbf{y})$ from training data

2) Compute output $\hat{\mathbf{y}}$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X}$$

3) Compute loss

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

4) Compute derivative

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$\odot$ is element-wise
multiplication

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m}$$

η is learning rate

More generalized formula

```

1 for epoch in range(epoch_max):
2     for i in range(0, N, m):
3         # get m samples
4         x = data[:, i:i+m]
5         y = prices[i:i+m, :]
6
7         # predict y_hat
8         y_hat = theta.T.dot(x)
9
10        # compute loss
11        loss = np.multiply((y_hat-y.T), (y_hat-y.T))
12        losses.append(np.mean(loss))
13
14        # compute gradient
15        k = 2*(y_hat-y.T)
16        gradients = np.multiply(np.vstack((k, k)), x)
17        gradients = gradients.dot(np.ones((m, 1))) / m
18
19        # update weights
20        theta = theta - lr*gradients

```

Linear Regression (m-samples)

Way 1.2

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}^{(i)} (\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$= \begin{bmatrix} -22.676 & -14.848 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} = \begin{bmatrix} -22.676 & -14.848 \\ -22.676 & -14.848 \end{bmatrix}$$

Take a look
and improve

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix}$$

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Linear Regression (m-samples)

Way 1.2

1) Pick m samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} = (\mathbf{x}^{(i)})^T \boldsymbol{\theta} \quad \text{for } 0 \leq i < m$$

3) Compute loss

$$L = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)})^2$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < m$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}}{m} \quad \eta \text{ is learning rate}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X} = [-2.238 \quad -1.524]$$

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$= \begin{bmatrix} -22.676 & -14.848 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 6.7 & 4.6 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} = \begin{bmatrix} -22.676 & -14.848 \\ -22.676 & -14.848 \end{bmatrix}$$

$$L'_{\boldsymbol{\theta}} = \mathbf{X} \mathbf{k}^T$$

$$\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix}$$

$$L'_{\boldsymbol{\theta}} = \left(\begin{bmatrix} \mathbf{k} \\ \mathbf{k} \end{bmatrix} \odot \mathbf{X} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -151.92 & -68.301 \\ -22.676 & -14.848 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Linear Regression

Way 1.2

1) Pick m samples $(\mathbf{x}, \mathbf{y})$ from training data

2) Compute output $\hat{\mathbf{y}}$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X}$$

3) Compute loss

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

4) Compute derivative

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$L'_{\boldsymbol{\theta}} = \mathbf{X} \mathbf{k}^T$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m}$$

η is learning rate

More generalized formula

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.genfromtxt('data.csv', delimiter=',')
5 areas = data[:,0]
6 prices = data[:,1:]
7 N = areas.size
8
9 # vector [x, b]^T
10 data = np.vstack([areas, np.ones((N,))])
11
12 # [w, b]
13 theta = np.array([[ -0.34],
14                  [ 0.04]])
15
16 # params
17 lr = 0.01
18 epoch_max = 1
19 m = 2
20
21 # logging
22 losses = []
```


Way 1.2

1) Pick m samples $(\mathbf{x}, \mathbf{y})$ from training data

2) Compute output $\hat{\mathbf{y}}$

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^T \mathbf{X}$$

3) Compute loss

$$L = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y}^T)(\hat{\mathbf{y}} - \mathbf{y}^T)^T$$

4) Compute derivative

$$\mathbf{k} = 2(\hat{\mathbf{y}} - \mathbf{y}^T)$$

$$L'_{\boldsymbol{\theta}} = \mathbf{X} \mathbf{k}^T$$

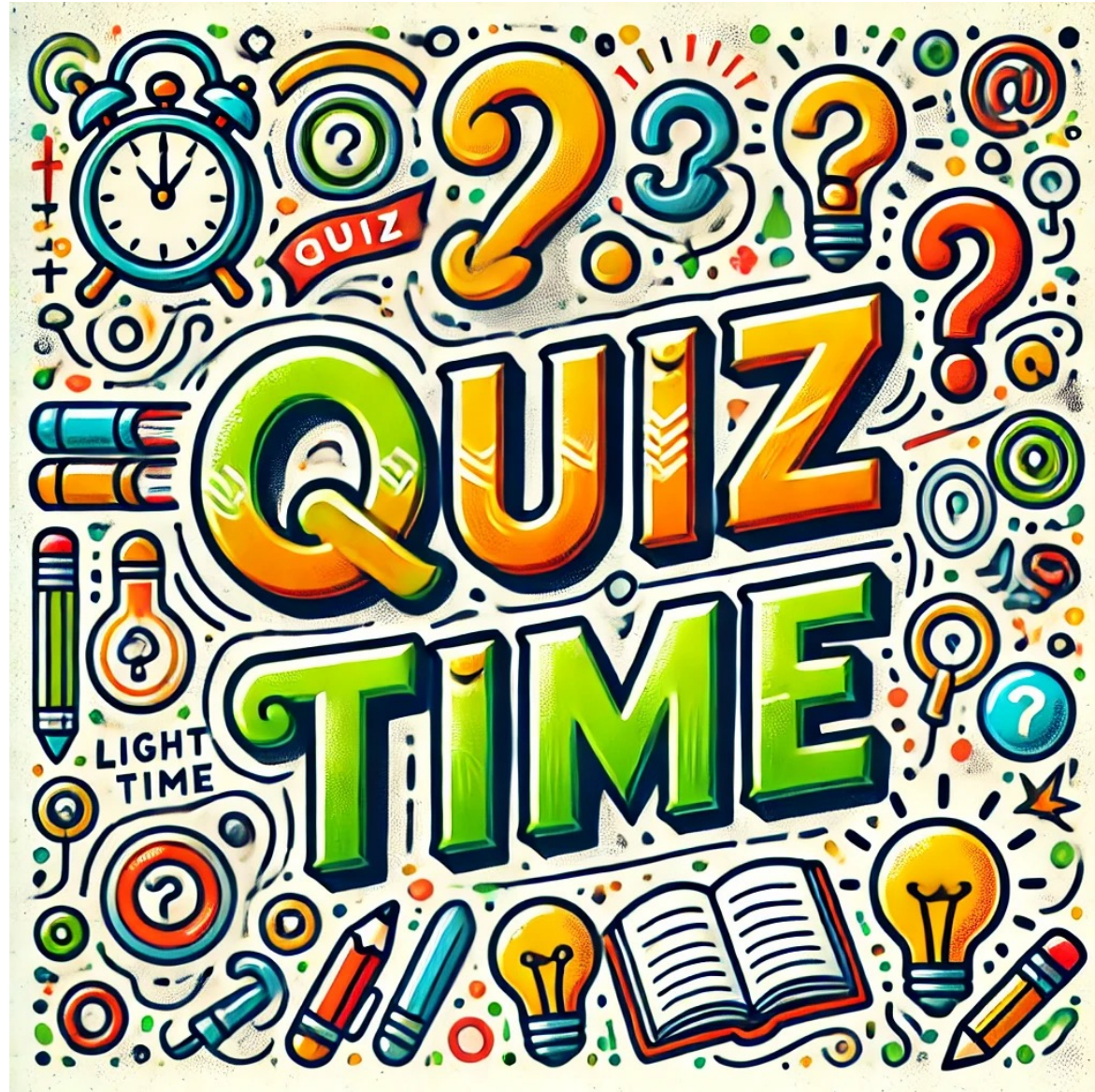
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{L'_{\boldsymbol{\theta}}}{m}$$

η is learning rate

More generalized formula

```
1 for epoch in range(epoch_max):
2     for i in range(0, N, m):
3         # get m samples
4         x = data[:, i:i+m]
5         y = prices[i:i+m, :]
6
7         # predict y_hat
8         y_hat = theta.T.dot(x)
9
10        # compute loss
11        loss = np.multiply((y_hat-y.T), (y_hat-y.T))
12        losses.append(np.mean(loss))
13
14        # compute gradient
15        k = 2*(y_hat-y.T)
16        gradients = x.dot(k.T) / m
17
18        # update weights
19        theta = theta - lr*gradients
```



❖ What about this arrangement?

Feature		Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$\hat{y} = X\theta = \begin{bmatrix} -2.238 \\ -1.524 \end{bmatrix}$$

$$L = \frac{1}{m}(\hat{y} - y)^T(\hat{y} - y) = \frac{1}{2}(128.5 + 55.11) = 91.8$$

$$k = 2(\hat{y} - y) = \begin{bmatrix} -22.676 \\ -14.848 \end{bmatrix}$$

$$L'_\theta = k^T X = \begin{bmatrix} -22.676 & -14.848 \end{bmatrix} \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix} = \begin{bmatrix} -220.23 & -37.524 \end{bmatrix}$$

$$\theta = \theta - \eta \left(\frac{L'_\theta}{m} \right)^T$$

$$= \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix} - 0.005 \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix} = \begin{bmatrix} 0.761 \\ 0.227 \end{bmatrix}$$

❖ What about this arrangement?

Feature		Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

1) Pick m samples (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

3) Compute loss

$$L = \frac{1}{m} (\hat{y} - y)^T (\hat{y} - y)$$

4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = k^T X$$

5) Update parameters

$$\theta = \theta - \eta \left(\frac{L'_{\theta}}{m} \right)^T \quad \eta \text{ is learning rate}$$

Linear Regression (m-samples)

34

❖ What about this arrangement?

Feature		Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

$$\hat{y} = X\theta = \begin{bmatrix} -2.238 \\ -1.524 \end{bmatrix}$$

$$L = \frac{1}{m} (\hat{y} - y)^T (\hat{y} - y)$$

$$k = 2(\hat{y} - y) = \begin{bmatrix} -22.676 \\ -14.848 \end{bmatrix}$$

$$L'_\theta = X^T k = \begin{bmatrix} 6.7 \\ 1 \end{bmatrix} \begin{bmatrix} 4.6 \\ 1 \end{bmatrix} \begin{bmatrix} -22.676 \\ -14.848 \end{bmatrix} = \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix}$$

$$\theta = \theta - \eta \frac{L'_\theta}{m}$$

$$= \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix} - 0.005 \begin{bmatrix} -220.23 \\ -37.524 \end{bmatrix} = \begin{bmatrix} 0.761 \\ 0.227 \end{bmatrix}$$

❖ What about this arrangement?

Feature		Label	
	area	price	
	6.7	9.1	
	4.6	5.9	
	3.5	4.6	
	5.5	6.7	

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \end{bmatrix}$$

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

1) Pick m samples (x, y) from training data

2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

3) Compute loss

$$L = \frac{1}{m}(\hat{y} - y)^T(\hat{y} - y)$$

4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = X^T k$$

5) Update parameters

$$\theta = \theta - \eta \frac{L'_{\theta}}{m}$$

η is learning rate

Outline

SECTION 1

1-sample Vectorization

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

SECTION 2

m-sample Vectorization

Way 1 for constructing matrix $\mathbf{x}$

$$\mathbf{x} = \begin{bmatrix} 6.7 & 4.6 & 3.5 & 5.5 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

SECTION 3

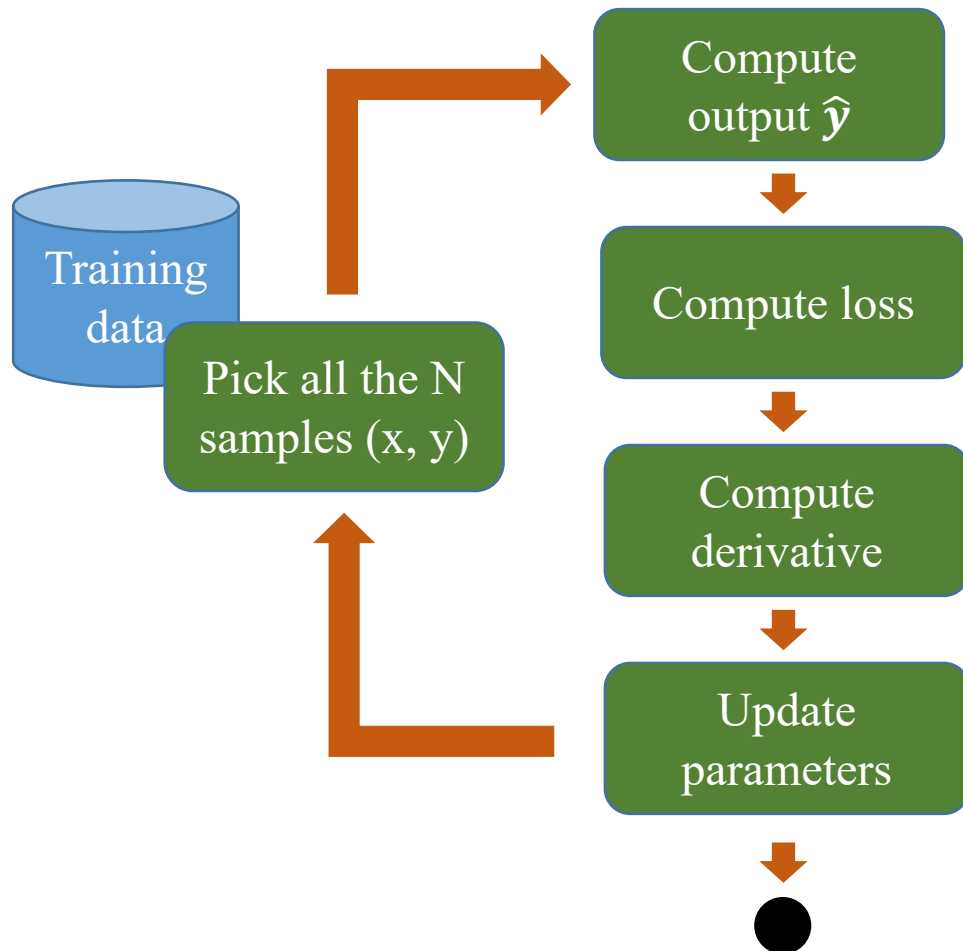
N-sample Vectorization

Way 2 for constructing matrix $\mathbf{x}$

$$\mathbf{x} = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \\ 3.5 & 1 \\ 5.5 & 1 \end{bmatrix}$$

❖ House price prediction

❖ N-sample training



1) Pick all the N samples $(x^{(i)}, y^{(i)})$ from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = wx^{(i)} + b \quad \text{for } 0 \leq i < N$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < N$$

4) Compute derivative

$$L'_w{}^{(i)} = 2x^{(i)}(\hat{y}^{(i)} - y^{(i)})$$
$$L'_b{}^{(i)} = 2(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < N$$

5) Update parameters

$$w = w - \eta \frac{\sum_i L'_w{}^{(i)}}{N}$$

$$b = b - \eta \frac{\sum_i L'_b{}^{(i)}}{N}$$

Learning rate η

1) Pick all the N samples from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = wx^{(i)} + b \quad \text{for } 0 \leq i < N$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < N$$

4) Compute derivative

$$\begin{aligned} L'_w{}^{(i)} &= 2x^{(i)}(\hat{y}^{(i)} - y^{(i)}) \\ L'_b{}^{(i)} &= 2(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < N \end{aligned}$$

5) Update parameters

$$\begin{aligned} w &= w - \eta \frac{\sum_i L'_w{}^{(i)}}{N} \\ b &= b - \eta \frac{\sum_i L'_b{}^{(i)}}{N} \end{aligned} \quad \eta \text{ is learning rate}$$

Friendly version

1) Pick all the N samples from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} \quad \text{for } 0 \leq i < N$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < N$$

4) Compute derivative

$$L'_\theta{}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < N$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_\theta{}^{(i)}}{N} \quad \eta \text{ is learning rate}$$

Generalized formula

1) Pick all the N samples from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} \quad \text{for } 0 \leq i < N$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < N$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < N$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}^{(i)}}{N} \quad \eta \text{ is learning rate}$$

Generalized formula

```
1 import numpy as np
2 from numpy import genfromtxt
3
4 data = genfromtxt('data.csv', delimiter=',')
5 areas = data[:,0]
6 prices = data[:,1]
7 data_size = areas.size
8
9 # vector [x, b]
10 data = np.c_[areas, np.ones((data_size, 1))]
11
12 n_epochs = 10
13 lr = 0.01
14
15 theta = np.array([[-0.34],[0.04]])
```

Linear Regression

1) Pick all the N samples from training data

2) Compute output $\hat{y}^{(i)}$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^T \mathbf{x}^{(i)} \quad \text{for } 0 \leq i < N$$

3) Compute loss

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{for } 0 \leq i < N$$

4) Compute derivative

$$L'_{\boldsymbol{\theta}}^{(i)} = 2\mathbf{x}^{(i)}(\hat{y}^{(i)} - y^{(i)}) \quad \text{for } 0 \leq i < N$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \frac{\sum_i L'_{\boldsymbol{\theta}}^{(i)}}{N} \quad \eta \text{ is learning rate}$$

Generalized formula

```
1 losses = [] # for debug
2 for epoch in range(n_epochs):
3     sum_of_losses = 0
4     gradients = np.zeros((2,1))
5
6     for index in range(data_size):
7         # get data
8         x_i = data[index:index+1]
9         y_i = prices[index:index+1]
10
11         # compute output y_hat_i
12         y_hat_i = x_i.dot(theta)
13
14         # compute loss
15         l_i = (y_hat_i - y_i)*(y_hat_i - y_i)
16
17         # compute gradient
18         g_l_i = 2*(y_hat_i - y_i)
19         gradient = x_i.T.dot(g_l_i)
20
21         # accumulate gradient
22         gradients = gradients + gradient
23         sum_of_losses = sum_of_losses + l_i
24
25     # normalize
26     sum_of_losses = sum_of_losses/data_size
27     gradients = gradients/data_size
28
29     # for debug
30     losses.append(sum_of_losses[0][0])
31
32     # update
33     theta = theta - lr*gradients
```

❖ Vectorization

House price data

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Model

$$\text{price} = w * \text{area} + b$$

$$\hat{y} = wx + b$$

$$y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

Parameter Initialization

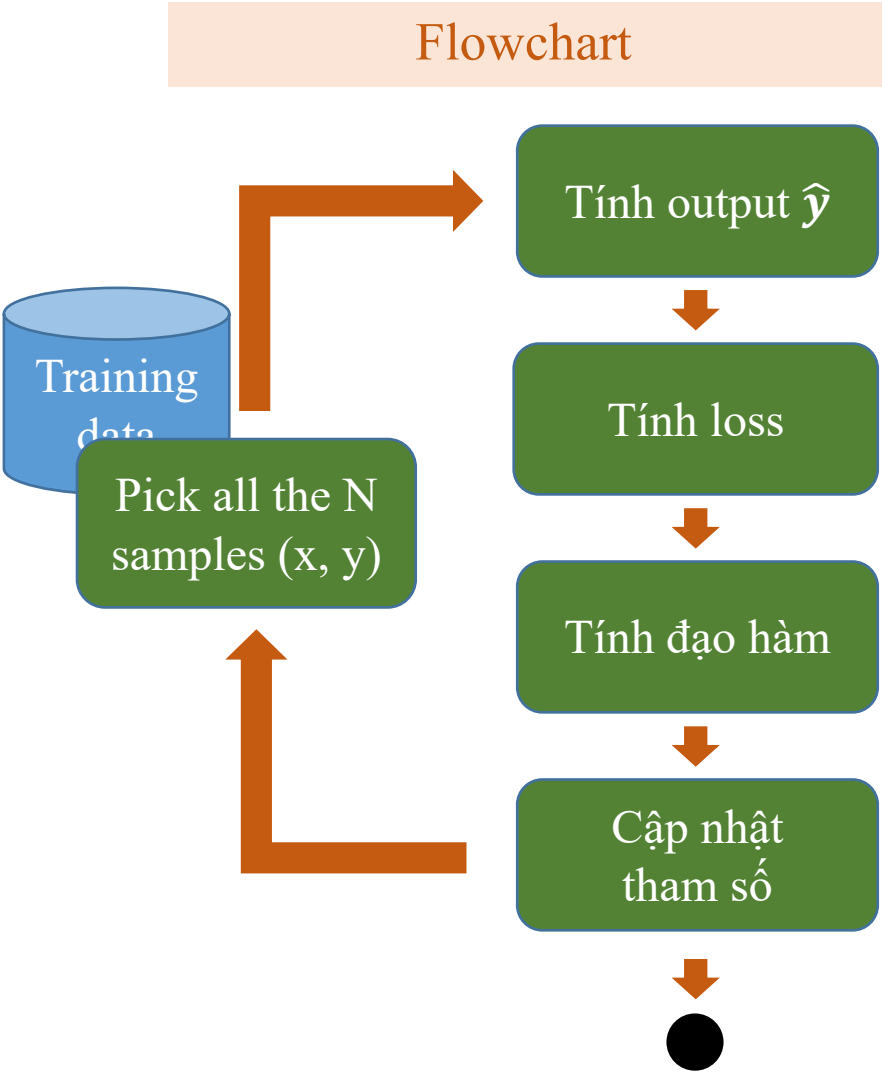
$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$

Way 1 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 4.6 & 3.5 & 5.5 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Way 2 for constructing matrix x

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \\ 3.5 & 1 \\ 5.5 & 1 \end{bmatrix}$$



Linear Regression [1]

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

House Price Data

Way 1

$$X = \begin{bmatrix} 6.7 & 4.6 & 3.5 & 5.5 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

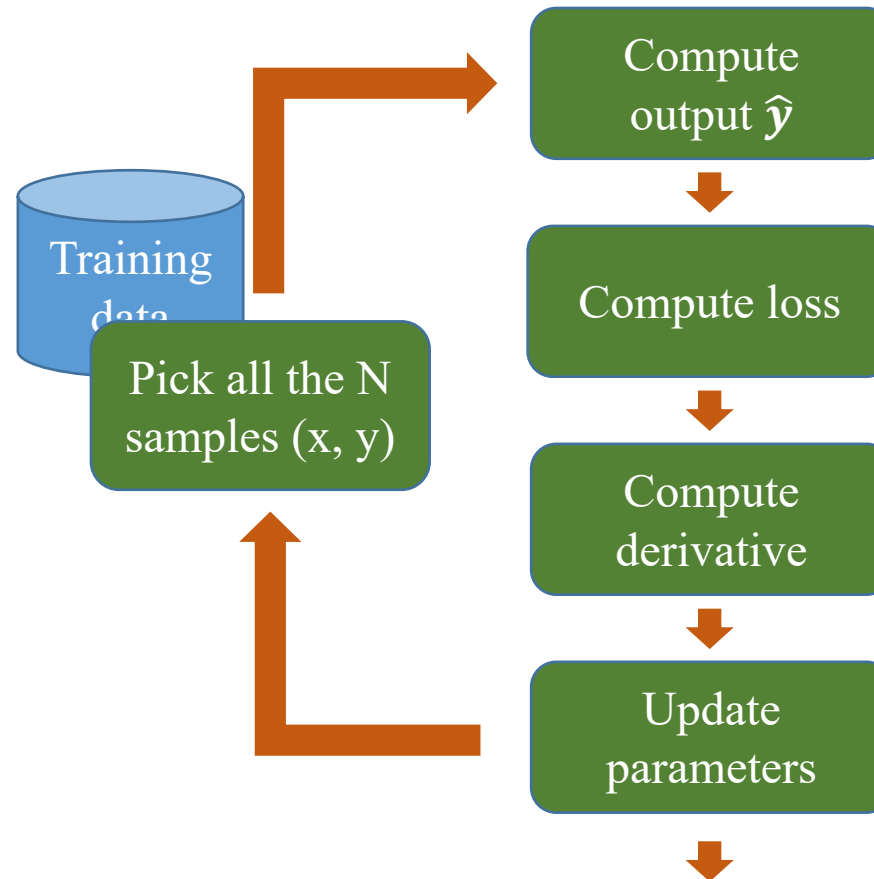
Model

$$\text{price} = w * \text{area} + b$$

$$\hat{y} = wx + b$$

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$



1) Pick all the N samples from training data

2) Compute output $\hat{y}$

$$\hat{y} = \theta^T X$$

3) Compute loss

$$L = \frac{1}{N} (\hat{y} - y)(\hat{y} - y)^T$$

4) Compute derivative

$$k = 2(\hat{y} - y^T)$$

$$L'_{\theta} = Xk^T$$

5) Update parameters

$$\theta = \theta - \eta \frac{L'_{\theta}}{N}$$

η is learning rate

Vectorization Approach

Linear Regression [2]

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

House Price Data

Way 2

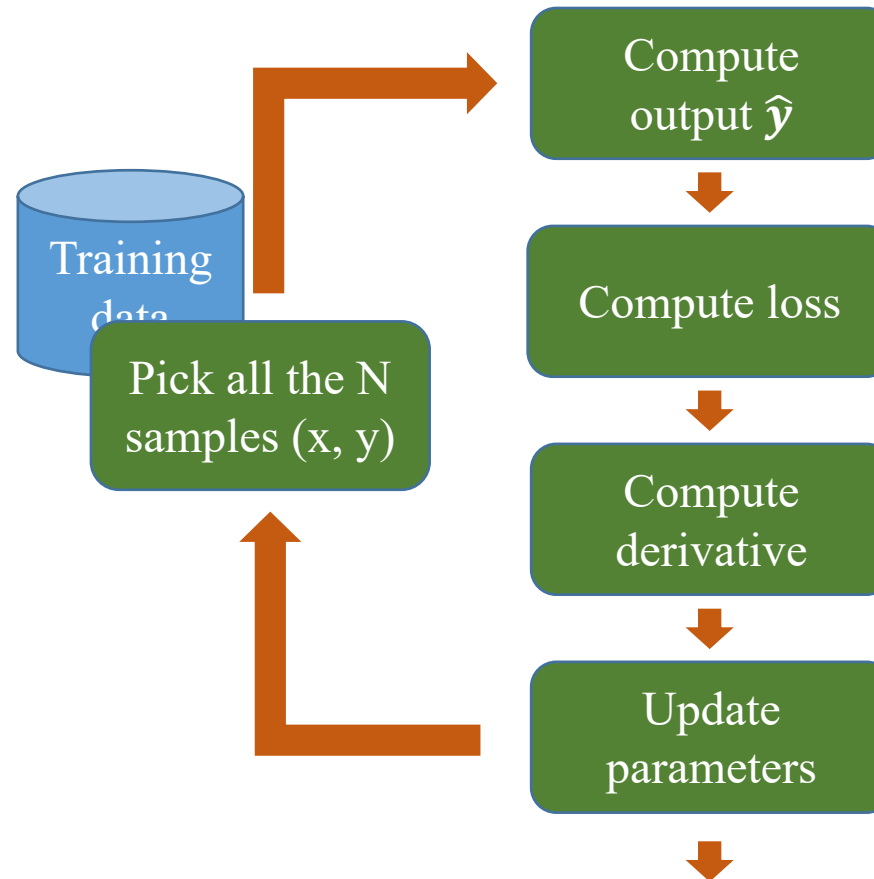
$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \\ 3.5 & 1 \\ 5.5 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

Model

$$\text{price} = w * \text{area} + b$$
$$\hat{y} = wx + b$$

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$



1) Pick all the N samples from training data

2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

3) Compute loss

$$L = \frac{1}{N} (\hat{y} - y)^T (\hat{y} - y)$$

4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = k^T X$$

5) Update parameters

$$\theta = \theta - \eta \left(\frac{L'_{\theta}}{N} \right)^T$$

η is learning rate

Vectorization Approach

Linear Regression [3]

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

House Price Data

Way 2

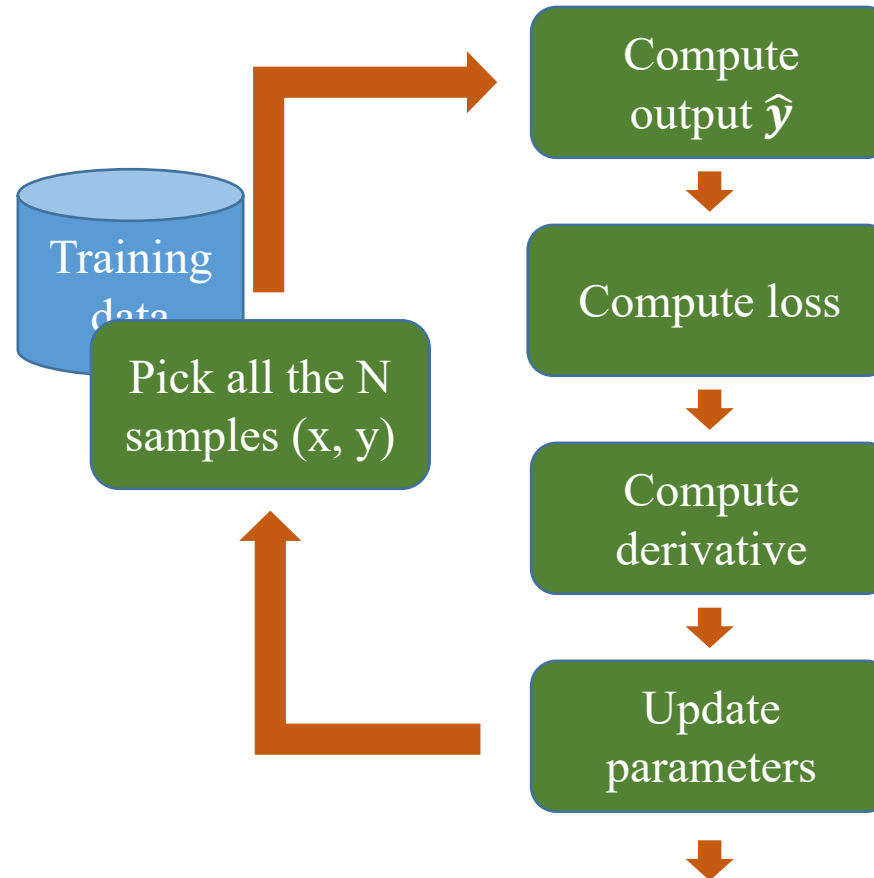
$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \\ 3.5 & 1 \\ 5.5 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

Model

$$\text{price} = w * \text{area} + b$$
$$\hat{y} = wx + b$$

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$



1) Pick all the N samples from training data

2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

3) Compute loss

$$L = \frac{1}{N} (\hat{y} - y)^T (\hat{y} - y)$$

4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = X^T k$$

5) Update parameters

$$\theta = \theta - \eta \frac{L'_{\theta}}{N}$$

η is learning rate

Vectorization Approach

Summary

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

House Price Data

Way 3

$$X = \begin{bmatrix} 6.7 & 1 \\ 4.6 & 1 \\ 3.5 & 1 \\ 5.5 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 9.1 \\ 5.9 \\ 4.6 \\ 6.7 \end{bmatrix}$$

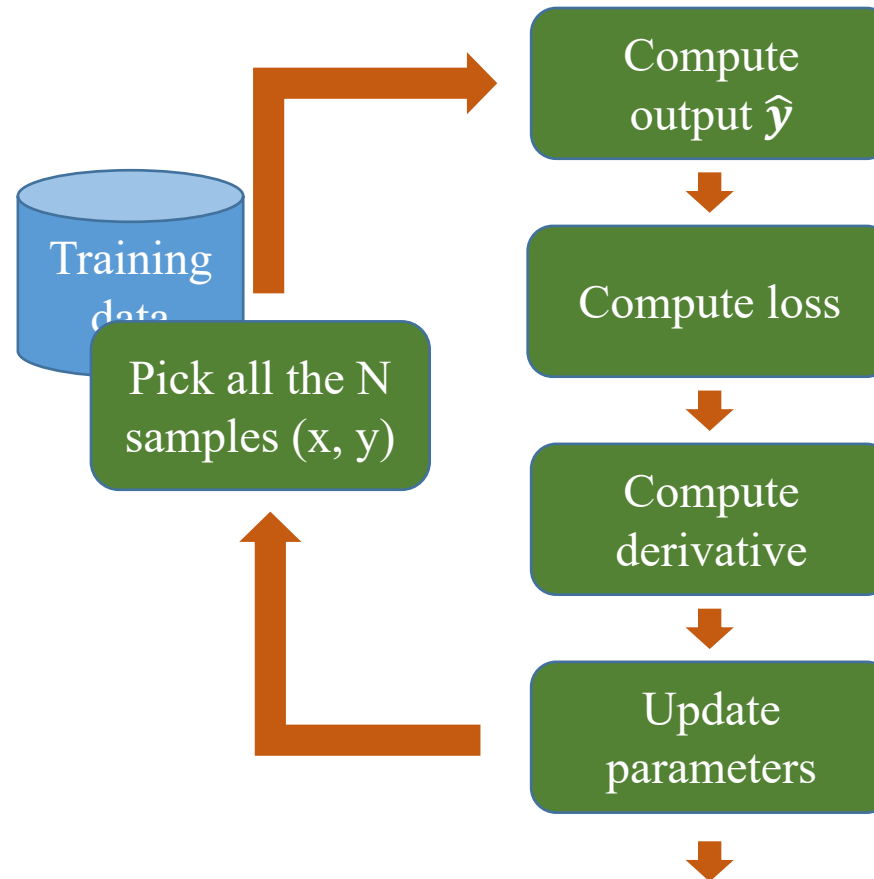
Model

$$\text{price} = w * \text{area} + b$$

$$\hat{y} = wx + b$$

Parameter Initialization

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} -0.34 \\ 0.049 \end{bmatrix}$$



- 1) Pick all the N samples from training data
- 2) Compute output $\hat{y}$

$$\hat{y} = X\theta$$

- 3) Compute loss

$$L = \frac{1}{N} (\hat{y} - y)^T (\hat{y} - y)$$

- 4) Compute derivative

$$k = 2(\hat{y} - y)$$

$$L'_{\theta} = X^T k$$

- 5) Update parameters

$$\theta = \theta - \eta \frac{L'_{\theta}}{N} \quad \eta \text{ is learning rate}$$

```
for epoch in range(epoch_max):
    y_hat = x.dot(theta)

    loss = np.multiply((y_hat-y), (y_hat-y))
    losses.append(np.mean(loss))

    k = 2*(y_hat-y)
    gradients = x.T.dot(k) / N

    theta = theta - lr*gradients
```